

Agentic AI Project Proposal

Problem Statement and Motivation:

Retail pharmacies are known to operate in fast-paced, high-volume environments. From customers waiting hours to get their prescription after being told it would be ready at a certain time, to waiting in line for hours only to be told their medicine is out of stock, couldn't go through because of insurance issues, or has refill problems. Burnt-out employees are also struggling to take vaccinations, help customers pick up medicine, fill prescriptions, give vaccines, and solve prescribing issues all at the same time. This high-stress environment in the retail pharmacy industry takes a toll on both pharmacy staff and customers, and it can be improved with the help of AI.

Implementing an agentic AI solution would be beneficial because pharmacy workflows involve multiple steps and decision points. Using an agentic AI solution can help autonomously retrieve and organize customer medication data, coordinate multiple specialized agents, and proactively surface relevant issues at the point of pickup. Using an agentic AI solution instead of a single AI model or static workflow helps ensure agents can plan, analyze, summarize, verify information, and adapt to different customer situations. Additionally, agentic AI with humans in the loop allows this approach to maintain safety and accountability.

Cases and Scenarios:

In scenario 1, a pharmacist is handling multiple tasks at once during peak hours and wants to view a customer's profile quickly. Once the pharmacist types in the customer's name in the AI system, they will be provided with the customer's profile and a brief AI overview of the customer's information in the top. The agent will be tasked with retrieving the customer profile,

summarizing active and recent medication, highlighting past issues such as missed refills, side effects, insurance delays etc. and will generate a concise pickup summary for the customer.

In scenario 2, a customer is picking up a medication similar to one they previously had issues with. After the pharmacy technician types in the customer's name into the AI system, the pharmacy staff gets notified. The agent's task is reviewing medication history for known issues, flagging potential concerns using conservative logic, recommending pharmacist consultation when appropriate, and creating a short alert summary for review.

In scenario 3, a customer needs to ask a common question about the medication they are picking up. The pharmacy tech can type in the situation into the ai agent and the agent will interpret the customers question, retrieve pre-approved, non diagnostic medication information, generate a patient friendly explanation, and indicate when a pharmacist involvement is required.

Initial System Design:

The preliminary list of agents and roles starts off with an intake agent, which identifies the customer and retrieves the customer profile and medication records. The medication history agent reviews past and current medications and identifies previously reported issues. The risk-flagging agent detects potential medication-related concerns and uses rule-based, conservative checks. The pharmacy technician assistant agent generates clear summaries for techs and suggests next steps or escalations. The compliance and safety agent ensures responses are non-diagnostic and applies pharmacy policy and safety language. Lastly, the critic and verification agent reviews outputs for accuracy, clarity, and completeness.

Some expected tools, APIs, and data sources include simulated customer profiles, a mock medication and issues history database, predefined pharmacy policy rules, local file storage, and a simple front-end dashboard for technician interaction.

Feasibility and Risks:

Several challenges may arise when implementing this AI system in a pharmacy setting. One major concern is AI hallucination of medication information, which could create safety risks for customers and potential liability for pharmacists. Additionally, the system may over-flag potential issues, leading to alert fatigue for both customers and pharmacy staff. Another significant challenge involves accessing customer data, as HIPAA regulations restrict the use of real patient health records, making it difficult to integrate authentic medical information into the AI system.

Project Timeline:

Jan 29 – Feb 4: Finalize project idea, system concept, and submit proposal.

Feb 5 – Feb 18: Refine system architecture using autonomy, reasoning, planning, memory, and state management concepts.

Feb 19 – Mar 4: Design data retrieval and tool usage using LLM agents with tools and retrieval-augmented generation.

Mar 5 – Mar 11: Build a simple front-end interface for pharmacy technician interaction (Milestone I due Mar 11).

Mar 12 – Mar 25: Implement coordinated multi-agent workflow and agent communication (Milestone II due Mar 25).

Mar 26 – Apr 1: Apply safety principles, create evaluation metrics, and conduct pilot testing (Milestone III due Apr 1).

Apr 2 – Apr 23: Refine the full system, prepare the final report and presentation, and deliver the capstone presentation with peer evaluation.